

QUEEN BEATRIX INTL, Aruba

WMO: 789820

Lat: 12.500N

Lon: 70.017W

Elev: 60

StdP: 14.66

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 00411

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	75.0	75.9	67.8	102.3	79.4	68.7	105.6	79.6	27.8	84.8	26.5	84.4	9.9	80	0.733

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	10.0	93.1	80.9	91.8	80.6	91.4	80.5	82.8	89.8	82.0	89.0	81.5	88.6	19.4	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.8	160.6	87.9	80.3	157.6	87.4	79.2	151.9	86.5	46.8	89.7	45.9	89.4	45.3	89.3	91.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.5	25.9	24.8	72.1	95.3	1.2	1.2	71.2	96.1	70.5	96.8	69.8	97.5	69.0	98.3
			68.9	84.5	1.3	2.2	68.0	86.1	67.2	87.4	66.5	88.6	65.6	90.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	84.2	81.5	81.8	82.7	84.0	85.0	85.5	85.2	86.4	86.5	85.2	84.0	82.4
	DBStd	2.25	1.46	1.32	1.43	1.37	1.33	1.26	1.26	1.31	1.73	2.03	1.92	1.72
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12478	978	891	1013	1020	1084	1064	1092	1127	1096	1090	1019	1004
	CDD65	7003	513	471	548	570	619	614	627	662	646	625	569	539
	CDH74	81541	5008	4652	5688	6410	7388	7526	7655	8422	8410	7856	6757	5769
	CDH80	31615	1228	1227	1690	2254	2967	3221	3217	3968	4114	3463	2586	1680
Wind	WSAvg	16.6	15.9	17.4	17.8	17.9	18.6	19.6	18.5	17.7	15.1	12.8	12.5	15.0
Precipitation	PrecAvg	16.4	1.7	0.8	0.4	0.6	0.5	0.9	1.2	0.7	1.5	2.2	3.4	2.5
	PrecMax	24.6	4.2	3.1	1.4	4.3	2.6	4.0	4.6	2.0	4.8	6.5	8.8	8.1
	PrecMin	7.8	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.6	0.3	0.3
	PrecStd	5.4	1.1	0.8	0.4	0.9	0.6	1.1	1.1	0.6	1.3	1.6	2.4	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.2	88.9	90.0	91.5	91.9	93.1	92.0	93.6	94.1	93.2	91.5	89.7
		MCWB	77.6	78.7	79.2	80.3	80.9	82.2	80.0	80.4	81.0	80.8	80.0	78.7
	2%	DB	87.5	87.9	88.6	89.9	91.2	91.5	91.4	92.6	93.1	91.6	89.9	88.1
		MCWB	77.4	77.7	78.3	79.4	80.8	81.1	80.1	80.4	81.1	80.5	79.7	78.3
	5%	DB	86.2	86.4	87.8	89.3	89.9	90.0	90.0	91.5	91.7	90.8	89.3	87.5
		MCWB	77.1	77.0	77.9	79.1	80.0	80.3	79.8	80.2	80.7	80.3	79.6	78.2
	10%	DB	85.7	86.0	86.6	88.0	89.2	89.5	89.5	90.6	91.2	89.5	88.0	86.2
		MCWB	76.9	76.7	77.3	78.7	79.7	80.0	79.7	79.9	80.6	80.1	79.5	77.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.1	80.7	81.3	81.9	83.7	83.8	82.7	82.8	83.5	82.7	82.1	80.1
		MCDB	84.8	86.3	88.9	88.1	90.3	90.7	89.1	90.5	91.5	89.5	88.0	86.6
	2%	WB	78.9	79.1	79.8	80.8	81.9	82.3	81.7	81.7	82.5	81.8	81.0	79.4
		MCDB	85.1	85.7	87.0	87.9	88.4	89.5	88.4	89.7	90.2	88.5	87.4	85.8
	5%	WB	78.1	78.1	78.9	80.0	81.0	81.3	80.9	81.0	81.7	81.1	80.3	78.9
		MCDB	84.4	85.0	85.9	87.1	87.8	88.4	87.9	88.9	89.2	87.8	86.8	85.3
	10%	WB	77.3	77.3	78.0	79.3	80.1	80.4	80.1	80.4	81.0	80.6	79.8	78.3
		MCDB	83.7	84.2	85.0	86.2	86.8	87.3	86.9	88.2	88.3	87.3	86.1	84.5
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.5	9.6	9.6	9.3	9.4	9.5	9.9	10.0	9.3	8.9	8.6
		MCDBR	9.3	9.8	10.0	10.2	9.8	9.7	10.0	10.4	10.5	9.9	9.4	9.1
		MCWBR	4.5	4.8	4.7	4.4	4.4	4.3	4.1	3.9	4.2	4.1	4.2	4.1
	5% WB	MCDBR	8.9	9.6	9.6	9.7	9.4	9.7	9.6	10.0	10.1	9.4	8.9	8.7
		MCWBR	5.0	5.2	5.1	4.8	4.8	4.8	4.6	4.2	4.6	4.5	4.6	4.6
Clear-Sky Solar Irradiance	taub	0.388	0.412	0.421	0.486	0.475	0.497	0.479	0.444	0.428	0.402	0.393	0.394	
	taud	2.542	2.447	2.435	2.233	2.301	2.229	2.291	2.426	2.454	2.561	2.576	2.553	
	Ebn at Noon	290	286	285	264	263	254	259	272	279	286	286	285	
	Edh at Noon	32	37	38	46	42	45	42	37	36	32	31	31	
All-Sky Solar Radiation	RadAvg	1690	1895	1994	2022	1989	2027	2072	2120	1992	1749	1596	1565	
	RadStd	66	52	79	63	79	70	53	65	113	107	156	82	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	-0.59	-0.43	-0.86	-0.86	N/A	N/A	N/A	N/A

Nomenclature: See separate page